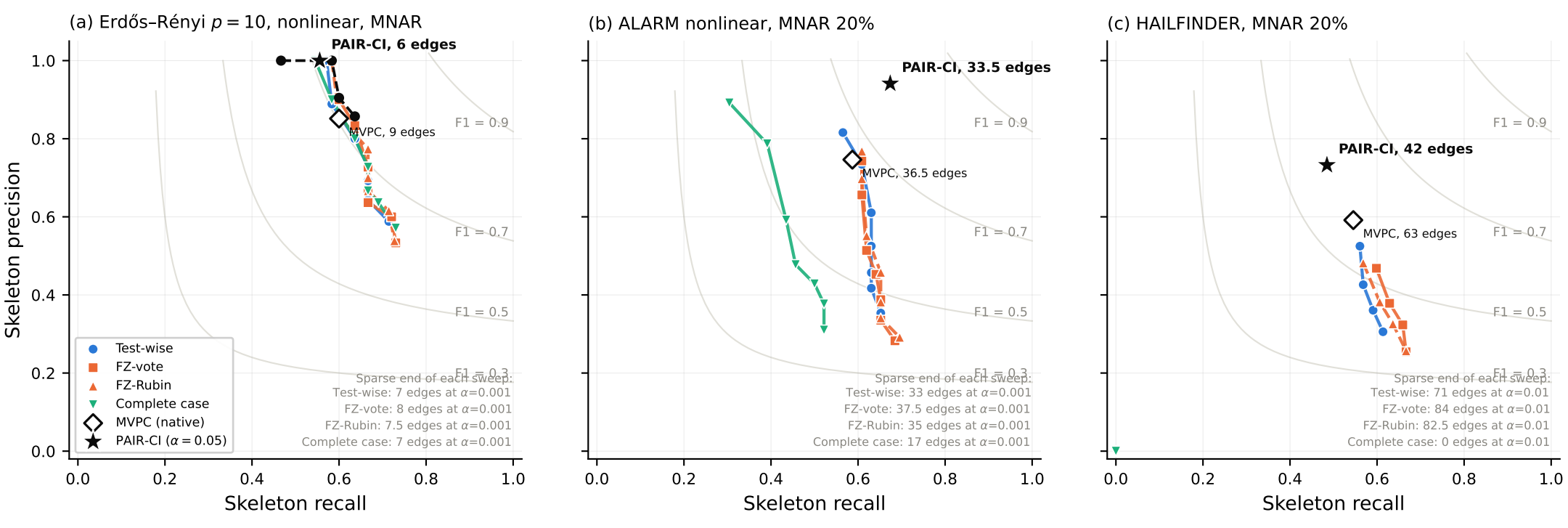
(a) Erdős–Rényi $p = 10$, nonlinear, MNAR
PAIR-CI, 6 edges
MVPC, 9 edges
F1 = 0.9
F1 = 0.7
F1 = 0.5
F1 = 0.3
Test-wise
FZ-vote
FZ-Rubin
Complete case
MVPC (native)
PAIR-CI ($\alpha = 0.05$)
Sparse end of each sweep:
Test-wise: 7 edges at α=0.001
FZ-vote: 8 edges at α=0.001
FZ-Rubin: 7.5 edges at α=0.001
Complete case: 7 edges at α=0.001
Skeleton precision
Skeleton recall
(b) ALARM nonlinear, MNAR 20%
PAIR-CI, 33.5 edges
MVPC, 36.5 edges
F1 = 0.9
F1 = 0.7
F1 = 0.5
F1 = 0.3
Sparse end of each sweep:
Test-wise: 33 edges at α=0.001
FZ-vote: 37.5 edges at α=0.001
FZ-Rubin: 35 edges at α=0.001
Complete case: 17 edges at α=0.001
Skeleton recall
(c) HAILFINDER, MNAR 20%
PAIR-CI, 42 edges
MVPC, 63 edges
F1 = 0.9
F1 = 0.7
F1 = 0.5
F1 = 0.3
Sparse end of each sweep:
Test-wise: 71 edges at α=0.01
FZ-vote: 84 edges at α=0.01
FZ-Rubin: 82.5 edges at α=0.01
Complete case: 0 edges at α=0.01
Skeleton recall